PHYSICAL CUMULATIVE EXAM

November 13, 1993

In 1921 the Faraday Society held a discussion on "The Radiation Theory of Chemical Action". At this meeting, the whole problem, including that of unimolecular reactions, was hotly debated. One who expressed strong opposition to the radiation hypothesis was Frederick Lindemann (later Lord Cherwell). Realizing that if he destroyed the hypothesis he should suggest an alternative explanation for unimolecular reactions, and he did so. The Lindemann hypothesis is the basis of all modern theories of unimolecular reactions, although a number of important modifications had to be made to it later.

The hypothesis provided a way, under some conditions, by which activation by collision can give rise to first-order kinetics. It also led to the important prediction that under certain conditions all unimolecular reactions must become second order. The Lindemann mechanism is

$$A + A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A^* + A$$

$$A^* \xrightarrow{k_2} Y + Z$$

where A^* is an energized molecule

1) Devise an equation for the rate of product formation (dY/dt).

2) Show what conditions are necessary for the observation of first-order kinetics and then show what conditions must prevail for the observation of second-order kinetics.

3) Show, mathematically, how one can determine the rate constant for the bimolecular activation (k_1).